Pratik Panwar | BS22B029

Indian Institute of Technology Madras

Pratik-Panwar | Pratik464iitm | pratik_cf464

Education

- **Indian Institute of Technology Madras** *Nov'22–May'27*
  Dual Degree in Biotechnology | CGPA: **7.94/10**

- **Shri Kanwartara Public Hr. Sec. School, Mandleshwar**
  Class 12 (CBSE): **92.6%** *May'21–Mar'22*

- **Shri Kanwartara Public Hr. Sec. School, Mandleshwar**
  Class 10 (CBSE): **91.0%** *May'19–Mar'20*

Scholastic Achievements

- Qualified **National Talent Search Examination (NTSE)**, placing in the **top 0.1%** among **1M+** candidates *(2020)*
- Ranked in the **top 1%** of **1.1M+** candidates in **JEE Main** and **top 4%** of **250K+** candidates in **JEE Advanced** *(2022)*

Professional Experience

Diax.ai

Software Engineering Intern *Mar'25–May'25*

- Built a full-stack healthcare platform using **React.js**, **Node.js**, **Express.js**, and **MongoDB** to streamline clinical workflows
- Developed **RESTful APIs** and designed MongoDB collections using Mongoose for efficient healthcare data management
- Automated publishing of **30+** blog posts through the **LinkedIn API**, using **Postman** for API testing and validation
- Built and deployed an automated transcription pipeline for **100+** doctor–patient consultations using **OpenAI Whisper**, streamlining clinical documentation and reducing manual effort

Coding Achievements

- Recognized among the top performers in **InMobi Codies 2026** Coding Contest, earning a **PPI** at **InMobi**
- Achieved a maximum **Codeforces** rating of **1667 (Expert)**
- Secured **Global Rank 386** in Codeforces Round 1102 (Div. 2) and **Global Rank 496** in Order Capital Round 2 (Div. 1 + 2)
- Achieved **Global Rank 101** in Starters 241 (CodeChef)
- Rated **1809 (Knight)** on **LeetCode**, placing in the **top 7.75%**

- Achieved **Global Rank 488** among **36K+** participants in the **ICPC 2025 Online Winter Challenge** powered by Huawei
- Secured **Rank 416** among **14K+** participants in the **Goldman Sachs India Hackathon 2026 – CS** challenge

- Ranked **18th** in **Reverse Coding Contest** (250+ participants)
- Ranked **11th** in **Heuristics Contest** held by Programming Club
- Ranked **12th** in **DSA Hackathon** held by Programming Club

Skills

Programming Languages: C++, C, Python, JavaScript, SQL
Web Technologies: HTML, CSS, React.js, Node.js, Express.js
Databases & Tools: MongoDB, Redis, Git, GitHub, Postman

Courses

- Data Structures & Algorithms
- Operating Systems*
- Computer Networks*
- Machine Learning Foundations
- Programming in Python
- Principles of Economics
- Web Development
- Object Oriented Programming
- Series and Matrices
- Money, Banking & Financial Markets

* : Ongoing

Projects

SyncBoard

Real-Time Collaborative Workspace *MERN, JWT, Socket.IO*

- Developed a **MERN-based** collaborative workspace with **RESTful APIs** for Workspaces, Boards, Columns, and Tasks, implementing complete **CRUD** functionality across modules
- Implemented **JWT authentication** and **RBAC** with workspace-level permissions for secure member collaboration
- Integrated **Socket.IO** for real-time synchronization and implemented **search, pagination, validation,** and **centralized error handling** for efficient and scalable collaboration

HuffZip

Lossless File Compression Utility *C++, STL, OOP, File I/O*

- Developed a **Huffman Coding**–based file compression system for **lossless compression** and decompression of text files using a **greedy algorithm** and **binary tree** encoding
- Implemented **Huffman Tree construction, code generation, bit packing,** and **binary file I/O** for efficient compression
- Designed a modular compression/decompression pipeline with separate encoding, decoding, and file-handling components

ChatRelay

Real-Time C++ Messaging Engine *Multithreading, Boost.Asio*

- Developed a **C++ client-server chat application** supporting real-time messaging among multiple users over **TCP sockets**
- Implemented **multithreaded, asynchronous client handling** using **Boost.Asio**, enabling concurrent user connections and efficient non-blocking message communication
- Designed a **structured messaging protocol** supporting private messaging, user renaming, and active client tracking for efficient message routing among connected clients

QuickLink

Scalable URL Shortening Service *MERN, JWT, Redis*

- Built a URL shortening service with **JWT authentication**, custom aliases, URL expiry, click analytics, and **RESTful APIs**
- Implemented **Redis caching**, **MongoDB indexing**, input validation, and secure backend architecture for URL retrieval
- Designed a **cache-aside architecture** to serve frequently accessed URLs from Redis and reduce database load

Positions of Responsibility

Alumni Relations Cell

Strategist *Apr'24–May'25*

- Led **Recall**, ARC's flagship mentorship initiative, hosting **10 alumni sessions** and managing a budget of **₹5+ lakhs**
- Coordinated an internship program achieving **100% placement**, facilitating **12 hires** from a pool of **500+** applicants

Teaching Assistant

GN6106 *May'23–Mar'24*

- Assisted with course coordination and academic activities for **90+ students** under **Prof. Sridhar Boovaraghavan**

Extra-Curriculars

Sports

- Selected for the **20-member Cricket NSO Contingent** at IIT Madras from a competitive pool of **200+** applicants
- Selected for the **IIT Madras Inter-IIT Cricket Contingent**
- Represented Brahmaputra (2024) and Godavari (2025–26) in the institute's **Inter-Hostel Cricket League** (Schorters)
- Secured **2nd Runner-up** in the Inter-Departmental Cricket League and was named **2nd Most Valuable Player**

- **SAATHI Mentor:** Mentored 3 freshmen, supporting their transition to college through peer engagement